

Maximizing Monotone Submodular Fns

We want to solve;

f a monotone submodular fn
 $M = (E, \mathcal{I})$ matroid

$$\begin{aligned} \max: & f(I) \\ \text{st: } & I \in \mathcal{I} \end{aligned}$$

Greedy: $1/2$ - approximation

Using multi linear extension, we get: $(1 - 1/e)$ -approx'n.

Roadmap

Step 1: We consider the problem

$$\begin{array}{|l} \max: F(x) \\ \text{st: } x \in P \end{array} \quad \left| \quad \begin{array}{l} \text{for } F \text{ } d^{\text{th}} \text{ multilinear} \\ \text{ext of a monotone} \\ \text{submodular } f. \end{array} \right.$$

P a polytope over which we can solve an LP

b/c f is submodular and hence F ~~increases~~ ^{increases} along $d \geq 0$

wlog assume P is down-monotone: $0 \leq y \leq x, x \in P \Rightarrow y \in P$.

We show: we can find $\hat{x} \in P$ with $F(\hat{x}) \geq (1-\epsilon) \text{OPT}$

Step 2: Given $\hat{x} \in P$, round to an ext point of P .

We will use $P = P(K)$, so we need to round $\hat{x}$ to $\hat{I} \in \mathcal{X}$.

We will show: $f(\hat{I}) \geq F(\hat{x})$ when f is submodular.

Maximizing the Multilinear Extension

We adopt a Frank-Wolfe strategy

$$x_{k+1} = x_k + \eta \cdot \underline{v_{\max}}$$

$$\left[\begin{array}{l} \text{where } v_{\max} \triangleq \arg \max_v: \nabla F(x_k)^T v \\ \text{st: } v \in P \end{array} \right] \text{LP.}$$

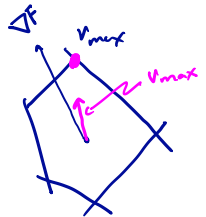
Continuous Greedy:

$$x(0) = 0.$$

$$\underline{v_{\max}}(x) = \arg \max_v: \nabla F(x)^T v \\ \text{st: } v \in P$$

$$\text{for } t \in [0, 1]: \quad \frac{dx}{dt} = v_{\max}(x(t))$$

Returns: $x(1)$.



Using Concavity

Lemma: For F multilinear extension of a monotone submodular function, and for $P \subseteq [0,1]^E$

$$\forall x \in P: \exists v \in P \text{ s.t. } \underline{\nabla F(x)^T v \geq \text{OPT} - F(x)}$$

pf. We know: $\nabla F(x) \geq 0$, also, F is concave
along any dir. $d \geq 0$:

$$\Rightarrow F(x+d) \leq F(x) + \nabla F(x)^T d$$

$$\Rightarrow \nabla F(x)^T d \geq F(x+d) - F(x)$$

$$\text{For: } d = (x^* - x) \vee 0$$

$$\nabla F(x)^T ((x^* - x) \vee 0) \geq F(x + (x^* - x) \vee 0) - F(x)$$

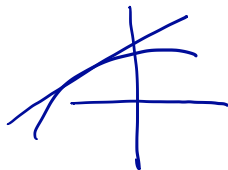
$$\geq F(x + x^* - x) - F(x) = \underline{\underline{F(x^*) - F(x)}}.$$

$$\text{Also: } \nabla F(x)^T \underline{x^*} \geq \nabla F(x)^T ((x^* - x) \vee 0) \text{ follows b/c } \nabla F \geq 0, \quad x^* \geq (x^* - x) \vee 0$$

Hence: $v = x^*$ satisfies req. of Lemma:

$$\nabla F(x)^T x^* \geq F(x^*) - F(x).$$

□.



Main Theorem

Thm. For F the multilinear
extension of a monotone $\uparrow$ submodular
fn f , $\forall$ Polytype $P \in [0, 1]^E$,

$$F(x(i)) \geq (1 - 1/e) \cdot OPT$$

where $x(i)$ output. of the Continuous Greedy
alg.

Proof of Main Theorem

$$\dot{x} = v_{\max}(x(t)), \quad x(0) = 0, \quad F(x(0)) = 0$$

$$x(1) = \int_0^1 v_{\max}(x(t)) dt \quad \text{Hence: } \underline{x(1) \in \mathbb{I}}$$

$$F(x(1)) = \int_0^1 \frac{d}{dt} F(x(t)) dt$$

$$\frac{d}{dt} F(x(t)) = \nabla F(x(t))^T \dot{x} = \nabla F(x(t))^T v_{\max}(x(t)) \geq \underline{\text{OPT} - F(x(t))}.$$

$$\frac{d}{dt} F(x(t)) + F(x(t)) \geq \text{OPT}.$$

$$\underline{\frac{d}{dt} (e^t F(x(t)))} = \underline{e^t \frac{d}{dt} F(x(t)) + e^t F(x(t))} \geq e^t \cdot \text{OPT}.$$

$$\begin{aligned} e^1 \cdot F(x(1)) &= \int_0^1 \frac{d}{dt} (e^t F(x(t))) dt \geq \int_0^1 (e^t \frac{d}{dt} F(x(t)) + e^t F(x(t))) dt \\ &\geq \int_0^1 e^t \cdot \text{OPT} \cdot dt = (e - 1) \cdot \text{OPT} \end{aligned}$$

$$\Rightarrow F(x(1)) \geq (1 - 1/e) \cdot \text{OPT}.$$



Back to the Agenda

This lecture concludes step 1.

$$F(x(i)) \geq^{(1-\frac{1}{e})} F(x^*) \geq F(z), \quad \forall z \in \underline{\text{Box}}$$
$$= f(z)$$

$$\text{Hence: } F(x(i)) \geq (1 - \frac{1}{e}) \cdot f(I^*)$$

We still have: step 2: How to get
from $x(i) \rightarrow I$.